

Applying these results, the system of six equations in Eq. (7.5.4) may be separated into two uncoupled systems, each containing three equations,

$$\begin{bmatrix} W/g & 0 & 0 \\ 0 & W/g & 0 \\ 0 & 0 & I_{yy_b} \end{bmatrix} \begin{Bmatrix} \Delta \dot{u} \\ \Delta \dot{v} \\ \Delta \dot{q} \end{Bmatrix} = \begin{Bmatrix} \Delta F_{x_b} + \Delta W_{x_b} \\ \Delta F_{z_b} + \Delta W_{z_b} + \Delta q V_o W/g \\ \Delta M_{y_b} \end{Bmatrix} \quad (7.5.15)$$

$$\begin{bmatrix} W/g & 0 & 0 \\ 0 & I_{xx_b} & -I_{xz_b} \\ 0 & -I_{xz_b} & I_{zz_b} \end{bmatrix} \begin{Bmatrix} \Delta \dot{v} \\ \Delta \dot{p} \\ \Delta \dot{r} \end{Bmatrix} = \begin{Bmatrix} \Delta F_{y_b} + \Delta W_{y_b} - \Delta r V_o W/g \\ \Delta M_{x_b} \\ \Delta M_{z_b} \end{Bmatrix} \quad (7.5.16)$$

or after using Eqs. (7.5.13) and (7.5.14),

$$\begin{aligned} & \begin{bmatrix} W/g - F_{x_b, \dot{u}} & -F_{x_b, \dot{v}} & 0 \\ -F_{z_b, \dot{u}} & W/g - F_{z_b, \dot{v}} & 0 \\ -M_{y_b, \dot{u}} & -M_{y_b, \dot{v}} & I_{yy_b} \end{bmatrix} \begin{Bmatrix} \Delta \dot{u} \\ \Delta \dot{v} \\ \Delta \dot{q} \end{Bmatrix} \\ &= \begin{bmatrix} F_{x_b, u} & F_{x_b, v} & F_{x_b, q} \\ F_{z_b, u} & F_{z_b, v} & F_{z_b, q} + V_o W/g \\ M_{y_b, u} & M_{y_b, v} & M_{y_b, q} \end{bmatrix} \begin{Bmatrix} \Delta u \\ \Delta v \\ \Delta q \end{Bmatrix} + \begin{bmatrix} F_{x_b, \delta_e} \\ F_{z_b, \delta_e} \\ M_{y_b, \delta_e} \end{bmatrix} \Delta \delta_e + \begin{Bmatrix} -W \cos(\theta_o) \\ -W \sin(\theta_o) \\ 0 \end{Bmatrix} \Delta \theta \end{aligned} \quad (7.5.17)$$

$$\begin{aligned} & \begin{bmatrix} W/g & 0 & 0 \\ 0 & I_{xx_b} & -I_{xz_b} \\ 0 & -I_{xz_b} & I_{zz_b} \end{bmatrix} \begin{Bmatrix} \Delta \dot{v} \\ \Delta \dot{p} \\ \Delta \dot{r} \end{Bmatrix} \\ &= \begin{bmatrix} F_{y_b, v} & F_{y_b, p} & F_{y_b, r} - V_o W/g \\ M_{x_b, v} & M_{x_b, p} & M_{x_b, r} \\ M_{z_b, v} & M_{z_b, p} & M_{z_b, r} \end{bmatrix} \begin{Bmatrix} \Delta v \\ \Delta p \\ \Delta r \end{Bmatrix} + \begin{bmatrix} F_{y_b, \delta_a} & F_{y_b, \delta_r} \\ M_{x_b, \delta_a} & M_{x_b, \delta_r} \\ M_{z_b, \delta_a} & M_{z_b, \delta_r} \end{bmatrix} \begin{Bmatrix} \Delta \delta_a \\ \Delta \delta_r \end{Bmatrix} \\ & \quad + \begin{Bmatrix} W \cos(\theta_o) \\ 0 \\ 0 \end{Bmatrix} \Delta \phi \end{aligned} \quad (7.5.18)$$

Using the small-disturbance variables from Eq. (7.5.3), applying the small-angle approximations,

$$\begin{aligned} \sin(\phi) &\cong \Delta \phi & \sin(\theta) &\cong \sin(\theta_o) + \cos(\theta_o) \Delta \theta & \sin(\psi) &\cong \Delta \psi \\ \cos(\phi) &\cong 1 & \cos(\theta) &\cong \cos(\theta_o) - \sin(\theta_o) \Delta \theta & \cos(\psi) &\cong 1 \end{aligned} \quad (7.5.19)$$

and neglecting second-order terms in the kinematic transformation matrix, Eq. (7.3.19) reduces to